Status	Finished
Started	Monday, 28 September 2026, 9:30 AM
Completed	Monday, 28 September 2026, 9:37 AM
Duration	7 mins 12 secs
Grade	**10.00** out of 10.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Given an array of integers `nums` containing `n + 1` integers where each integer is in the range `[1, n]` inclusive.There is only **one repeated number** in `nums`, return *this repeated number*. Solve the problem using set.

Example 1:

```
Input: nums = [1,3,4,2,2]
```

```
Output: 2
```

Example 2:

```
Input: nums = [3,1,3,4,2]
```

```
Output: 3
```

For example:

Input	Result
1 3 4 4 2	4

Answer: (penalty regime: 0 %)

```
import sys

def find_duplicate(nums):
    seen = set()
    for num in nums:
        if num in seen:
            return num
        seen.add(num)
    return -1

def main():
    input_data = sys.stdin.read().split()
    if not input_data:
        return

    nums = [int(x) for x in input_data]
    print(find_duplicate(nums))

if __name__ == '__main__':
    main()
```

	Input	Expected	Got	
✓	1 3 4 4 2	4	4	✓
✓	1 2 2 3 4 5 6 7	2	2	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **2**

Correct

Mark 1.00 out of 1.00

Check if a set is a subset of another set.

Example:

Sample Input1:

mango apple

mango orange

mango

output1:

yes

set3 is subset of set1 and set2

```
input2:
```

```
mango orange
```

```
banana orange
```

```
grapes
```

```
output2:
```

no

For example:

Test	Input	Result
1	mango apple mango orange mango	yes set3 is subset of set1 and set2
2	mango orange banana orange grapes	No

Answer: (penalty regime: 0 %)

```
import sys

def main():
    lines = sys.stdin.read().splitlines()
    if len(lines) < 3:
        return

    set1 = set(lines[0].split())
    set2 = set(lines[1].split())
    set3 = set(lines[2].split())

    if set3.issubset(set1) and set3.issubset(set2):
        print("yes")
        print("set3 is subset of set1 and set2")
    else:
        print("No")

if __name__ == '__main__':
    main()
```

	Test	Input	Expected	Got	
✓	1	mango apple mango orange mango	yes set3 is subset of set1 and set2	yes set3 is subset of set1 and set2	✓
✓	2	mango orange banana orange grapes	No	No	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **3**

Correct

Mark 1.00 out of 1.00

There is a malfunctioning keyboard where some letter keys do not work. All other keys on the keyboard work properly.

Given a string text of words separated by a single space (no leading or trailing spaces) and a string brokenLetters of all distinct letter keys that are broken, return the number of words in text you can fully type using this keyboard.

Example 1:

Input: text = "hello world", brokenLetters = "ad"

Output:

1

Explanation: We cannot type "world" because the 'd' key is broken.

For example:

Input	Result
hello world ad	1
Faculty Upskilling in Python Programming ak	2

Answer: (penalty regime: 0 %)

```
import sys

def main():
    lines = sys.stdin.read().splitlines()
    if not lines:
        return

    text = lines[0]
    brokenLetters = lines[1] if len(lines) > 1 else ""

    # Convert broken letters to lowercase set for case-insensitive checking
    broken_set = set(brokenLetters.lower())
    words = text.split()
    count = 0

    for word in words:
        # Check if any character (lowercased) is in broken_set
        if not any(char.lower() in broken_set for char in word):
            count += 1

    print(count)

if __name__ == '__main__':
    main()
```

	Input	Expected	Got	
✓	hello world ad	1	1	✓
✓	Welcome to REC e	1	1	✓
✓	Faculty Upskilling in Python Programming ak	2	2	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **4**

Correct

Mark 1.00 out of 1.00

Write a program to eliminate the common elements in the given 2 arrays and print only the non-repeating elements and the total number of such non-repeating elements.

Input Format:

The first line contains space-separated values, denoting the size of the two arrays in integer format respectively.

The next two lines contain the space-separated integer arrays to be compared.

Sample Input:

5 4

1 2 8 6 5

2 6 8 10

Sample Output:

1 5 10

3

Sample Input:

5 5

1 2 3 4 5

1 2 3 4 5

Sample Output:

NO SUCH ELEMENTS

For example:

Input	Result
5 4 1 2 8 6 5 2 6 8 10	1 5 10 3
5 5 1 2 3 4 5 1 2 3 4 5	NO SUCH ELEMENTS

Answer: (penalty regime: 0 %)

```
import sys

def main():
    input_data = sys.stdin.read().splitlines()
    if not input_data:
        return

    # Parse inputs
    arr1 = list(map(int, input_data[1].split()))
    arr2 = list(map(int, input_data[2].split()))

    set1 = set(arr1)
    set2 = set(arr2)

    # Find non-common elements while preserving original input order
    non_repeating = []

    # Add elements from arr1 that are not in arr2
    for x in arr1:
        if x not in set2 and x not in non_repeating:
            non_repeating.append(x)

    # Add elements from arr2 that are not in arr1
    for x in arr2:
        if x not in set1 and x not in non_repeating:
            non_repeating.append(x)

    # Output result
    if not non_repeating:
        print("NO SUCH ELEMENTS")
    else:
        print(*(non_repeating))
        print(len(non_repeating))

if __name__ == '__main__':
    main()
```

	Input	Expected	Got	
✓	5 4 1 2 8 6 5 2 6 8 10	1 5 10 3	1 5 10 3	✓
✓	3 3 10 10 10 10 11 12	11 12 2	11 12 2	✓
✓	5 5 1 2 3 4 5 1 2 3 4 5	NO SUCH ELEMENTS	NO SUCH ELEMENTS	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **5**

Correct

Mark 1.00 out of 1.00

Program to print all the distinct elements in an array. Distinct elements are nothing but the unique (non-duplicate) elements present in the given array.

Input Format:

First line take an Integer input from stdin which is array length n.

Second line take n Integers which is inputs of array.

Output Format:

Print the Distinct Elements in Array in single line which is space Separated

Example Input:

5

1 2 2 3 4

Output:

1 2 3 4

Example Input:

6

1 1 2 2 3 3

Output:

1 2 3

For example:

Input	Result
5 1 2 2 3 4	1 2 3 4

Answer: (penalty regime: 0 %)

```
import sys

def main():
    input_data = sys.stdin.read().split()
    if not input_data:
        return

    n = int(input_data[0])
    nums = [int(x) for x in input_data[1:n+1]]

    seen = set()
    distinct_elements = []

    for num in nums:
        if num not in seen:
            seen.add(num)
            distinct_elements.append(num)

    print(*(distinct_elements))

if __name__ == '__main__':
    main()
```

	Input	Expected	Got	
✓	5 1 2 2 3 4	1 2 3 4	1 2 3 4	✓

	Input	Expected	Got	
✓	6 1 1 2 2 3 3	1 2 3	1 2 3	✓
✓	5 11 22 11 22 11	11 22	11 22	✓
✓	10 1 2 3 4 5 1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **6**

Correct

Mark 1.00 out of 1.00

Given a tuple and a positive integer k, the task is to find the count of distinct pairs in the tuple whose sum is equal to **K**.

Examples:

> **Input:** t = (5, 6, 5, 7, 7, 8), K = 13
> **Output:** 2
> **Explanation:**
> Pairs with sum K(= 13) are {(5, 8), (6, 7), (6, 7)}.
> Therefore, distinct pairs with sum K(= 13) are { (5, 8), (6, 7) }.
> Therefore, the required output is 2.

For example:

Input	Result
1,2,1,2,5 3	1
1,2 0	0

Answer: (penalty regime: 0 %)

```
import sys

def main():
    lines = sys.stdin.read().splitlines()
    if not lines:
        return

    # Handle comma-separated or space-separated tuple input
    first_line = lines[0].strip()
    if ',' in first_line:
        t = [int(x) for x in first_line.split(',') if x.strip()]
    else:
        t = [int(x) for x in first_line.split()]

    K = int(lines[1].strip())

    seen_elements = set()
    unique_pairs = set()

    for x in t:
        complement = K - x
        if complement in seen_elements:
            # Store pair in sorted order so (a, b) is identical to (b, a)
            pair = (min(x, complement), max(x, complement))
            unique_pairs.add(pair)
        seen_elements.add(x)

    print(len(unique_pairs))

if __name__ == '__main__':
    main()
```

	Input	Expected	Got	
✓	5,6,5,7,7,8 13	2	2	✓
✓	1,2,1,2,5 3	1	1	✓
✓	1,2 0	0	0	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **7**

Correct

Mark 1.00 out of 1.00

You are given an integer tuple `nums` containing distinct numbers. Your task is to perform a sequence of operations on this tuple until it becomes empty. The operations are defined as follows:

1. If the first element of the tuple has the smallest value in the entire tuple, remove it.
2. Otherwise, move the first element to the end of the tuple.

You need to return an integer denoting the number of operations required to make the tuple empty.

Constraints

- The input tuple `nums` contains distinct integers.
- The operations must be performed using tuples and sets to maintain immutability and efficiency.
- Your function should accept the tuple `nums` as input and return the total number of operations as an integer.

Example:

Input: nums = (3, 4, -1)
Output: 5

Explanation:
Operation 1: [3, 4, -1] -> First element is not the smallest, move to the end -> [4, -1, 3]
Operation 2: [4, -1, 3] -> First element is not the smallest, move to the end -> [-1, 3, 4]
Operation 3: [-1, 3, 4] -> First element is the smallest, remove it -> [3, 4]
Operation 4: [3, 4] -> First element is the smallest, remove it -> [4]
Operation 5: [4] -> First element is the smallest, remove it -> []
Total operations: 5

For example:

Test	Result
`print(count_operations((3, 4, -1)))`	5

Answer: (penalty regime: 0 %)

Reset answer

```
def count_operations(nums: tuple) -> int:
    operations = 0

    # Continue until the tuple is empty
    while nums:
        operations += 1
        current_min = min(nums)

        if nums[0] == current_min:
            # Remove the smallest element
            nums = nums[1:]
        else:
            # Move the first element to the end
            nums = nums[1:] + (nums[0],)

    return operations

# Driver code to handle input format if called standalone
if __name__ == '__main__':
    import sys
    input_text = sys.stdin.read().strip()
    if input_text:
        # Parse tuple representation e.g. "(3, 4, -1)" or "3, 4, -1"
        cleaned = input_text.replace('(', '').replace(')', '')
        nums = tuple(map(int, cleaned.split(',')))
        print(count_operations(nums))
```

	Test	Expected	Got	
✓	`print(count_operations((3, 4, -1)))`	5	5	✓
✓	`print(count_operations((1, 2, 3, 4, 5)))`	5	5	✓
✓	`print(count_operations((5, 4, 3, 2, 1)))`	15	15	✓
✓	`print(count_operations((42, )))`	1	1	✓
✓	`print(count_operations((-2, 3, -5, 4, 1)))`	11	11	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **8**

Correct

Mark 1.00 out of 1.00

Coders here is a simple task for you, Given string str. Your task is to check whether it is a binary string or not by using python set.

Examples:

Input: str = "01010101010"

Output: Yes

Input: str = "REC101"

Output: No

For example:

Input	Result
01010101010	Yes
010101 10101	No

Answer: (penalty regime: 0 %)

```
import sys

def is_binary_string(s: str) -> str:
    # Convert string characters into a set
    char_set = set(s)

    # Valid binary characters allowed
    binary_set = {'0', '1'}

    # Check if char_set is a subset of {'0', '1'}
    if char_set.issubset(binary_set):
        return "Yes"
    else:
        return "No"

def main():
    # Read entire input string including any space or newline
    input_data = sys.stdin.read().splitlines()
    if not input_data:
        return

    s = input_data[0]
    print(is_binary_string(s))

if __name__ == '__main__':
    main()
```

	Input	Expected	Got	
✓	01010101010	Yes	Yes	✓
✓	REC123	No	No	✓
✓	010101 10101	No	No	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **9**

Correct

Mark 1.00 out of 1.00

The **DNA sequence** is composed of a series of nucleotides abbreviated as `'A'`, `'C'`, `'G'`, and `'T'`.

- For example, `"ACGAATTCCG"` is a **DNA sequence**.

When studying **DNA**, it is useful to identify repeated sequences within the DNA.

Given a string `s` that represents a **DNA sequence**, return all the `10`**-letter-long** sequences (substrings) that occur more than once in a DNA molecule. You may return the answer in **any order**.

Example 1:

```
Input: s = "AAAAACCCCCAAAAACCCCCCAAAAAGGGTTT"
Output: ["AAAAACCCCC","CCCCCAAAAA"]
```

Example 2:

```
Input: s = "AAAAAAAAAAAAA"
Output: ["AAAAAAAAAA"]
```

For example:

Input	Result
AAAAACCCCCAAAAACCCCCCAAAAAGGGTTT	AAAAACCCCC CCCCCAAAAA

Answer: (penalty regime: 0 %)

```
import sys

def find_repeated_dna_sequences(s: str):
    seen = set()
    repeated_set = set()
    result = []

    for i in range(len(s) - 9):
        sub = s[i:i+10]
        if sub in seen:
            if sub not in repeated_set:
                repeated_set.add(sub)
                result.append(sub)
        else:
            seen.add(sub)

    return result

def main():
    lines = sys.stdin.read().splitlines()
    if not lines or not lines[0].strip():
        return

    s = lines[0].strip()
    result = find_repeated_dna_sequences(s)

    for seq in result:
        print(seq)

if __name__ == '__main__':
    main()
```

	Input	Expected	Got	
✓	AAAAACCCCCAAAAACCCCCCAAAAAGGGTTT	AAAAACCCCC CCCCCAAAAA	AAAAACCCCC CCCCCAAAAA	✓
✓	AAAAAAAAAAAAA	AAAAAAAAAA	AAAAAAAAAA	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **10**

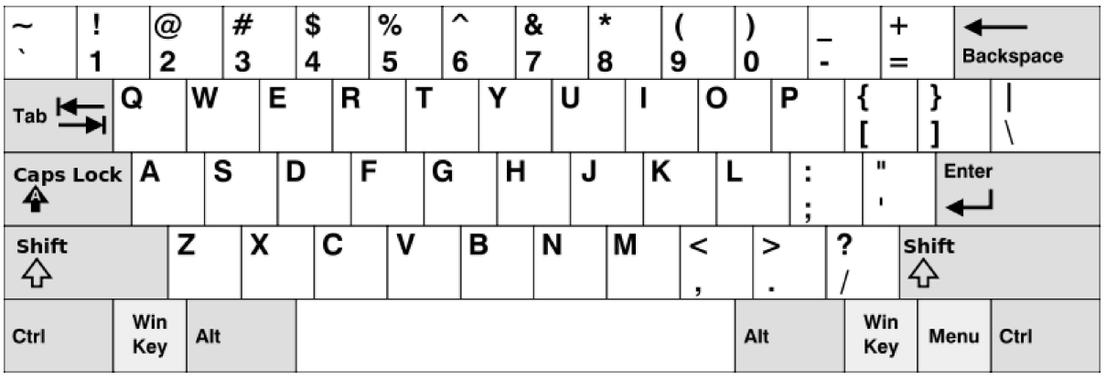
Correct

Mark 1.00 out of 1.00

Given an array of strings `words`, return *the words that can be typed using letters of the alphabet on only one row of American keyboard like the image below.*

In the **American keyboard**:

- the first row consists of the characters `"qwertyuiop"`,
- the second row consists of the characters `"asdfghjkl"`, and
- the third row consists of the characters `"zxcvbnm"`.



Example 1:

```
Input: words = ["Hello","Alaska","Dad","Peace"]
Output: ["Alaska","Dad"]
```

Example 2:

```
Input: words = ["omk"]
Output: []
```

Example 3:

```
Input: words = ["adsdf","sfd"]
Output: ["adsdf","sfd"]
```

For example:

Input	Result
4 Hello Alaska Dad Peace	Alaska Dad
2 adsfd afd	adsfd afd

Answer: (penalty regime: 0 %)

```
import sys

def main():
    lines = sys.stdin.read().splitlines()
    if not lines or not lines[0].strip():
        return

    n = int(lines[0].strip())
    words = [lines[i].strip() for i in range(1, n + 1) if i < len(lines)]

    row1 = set("qwertyuiop")
    row2 = set("asdfghjkl")
    row3 = set("zxcvbnm")

    result = []
    for word in words:
        w_set = set(word.lower())
        if w_set.issubset(row1) or w_set.issubset(row2) or w_set.issubset(row3):
            result.append(word)

    if not result:
        print("No words")
    else:
        for word in result:
            print(word)

if __name__ == '__main__':
```

```
27 ▾ if __name__ == "__main__":
28       main()
```

	Input	Expected	Got	
✓	4 Hello Alaska Dad Peace	Alaska Dad	Alaska Dad	✓
✓	1 omk	No words	No words	✓
✓	2 adsfd afd	adsfd afd	adsfd afd	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.